

Partial Differentiation

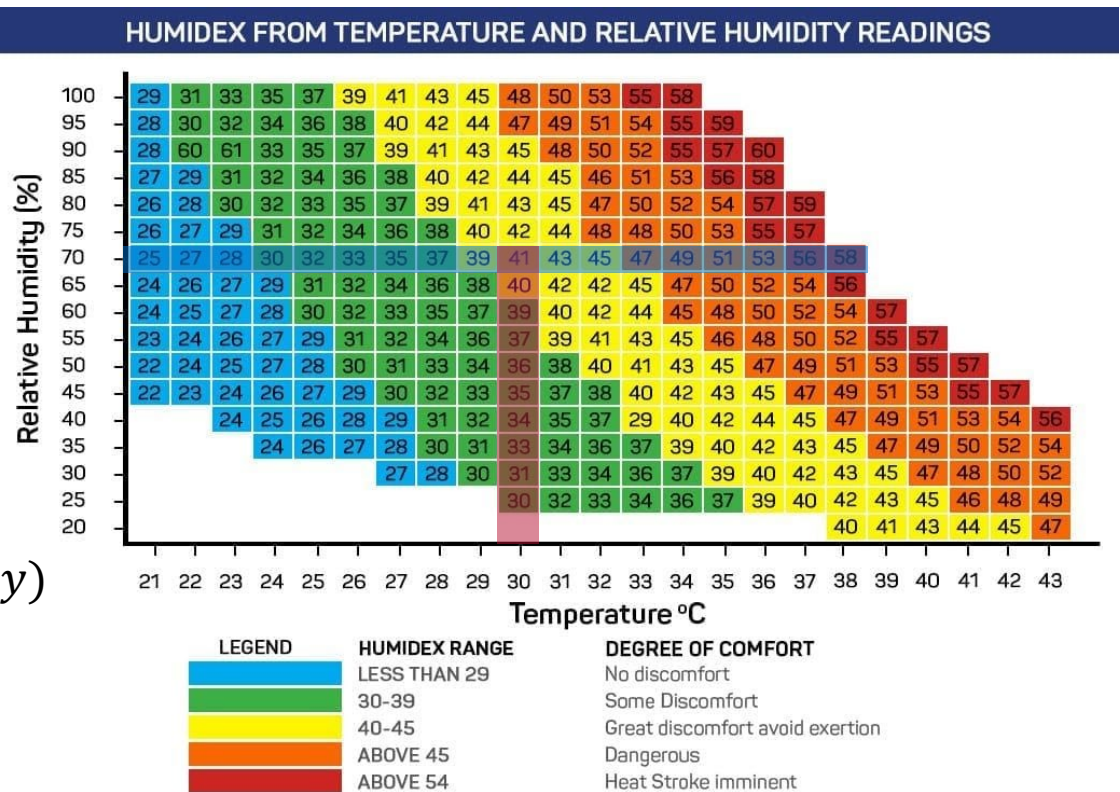
Trim Ch 11, S. 12.3

Outline:

1. Partial differentiation of $f(x, y)$
2. Notation (∂)



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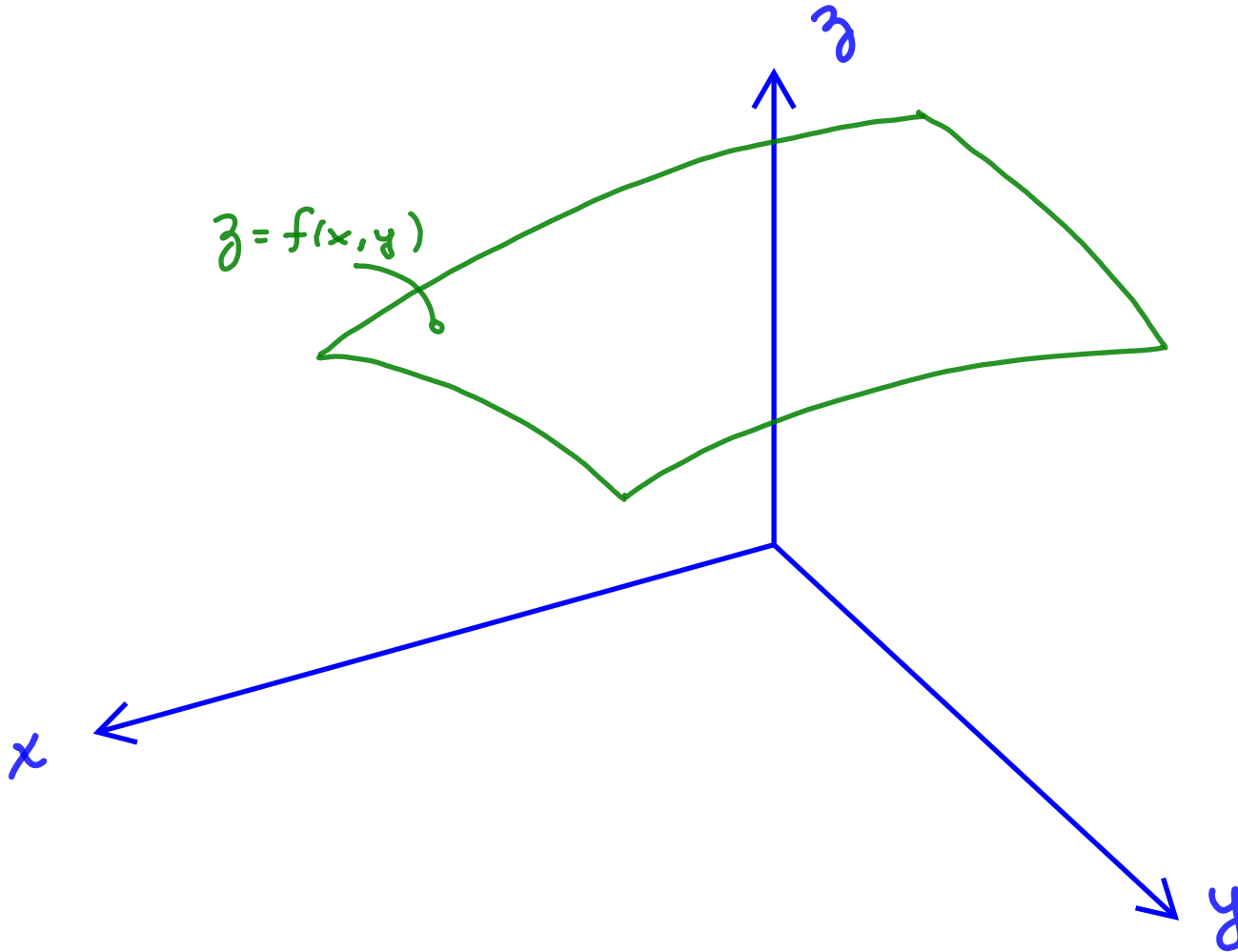
Source: Environment Canada

<https://www.canada.ca/en/environment-climate-change/services/seasonal-weather-hazards/spring-summer.html#humidex>

Source: https://www.ccohs.ca/oshanswers/phys_agents/humidex.html

Partial Derivative of a Function of Two Variables

When we hold all but one of the independent variables of a function constant and differentiate with respect to it, we get a partial derivative.



Partial Derivative of a Function of Two Variables

The slope of the tangent line at P is the rate of change of $f(x, y)$ as we vary x , holding y constant.

This is known as the first order partial derivative and provided that the limit exists, it can be calculated as,

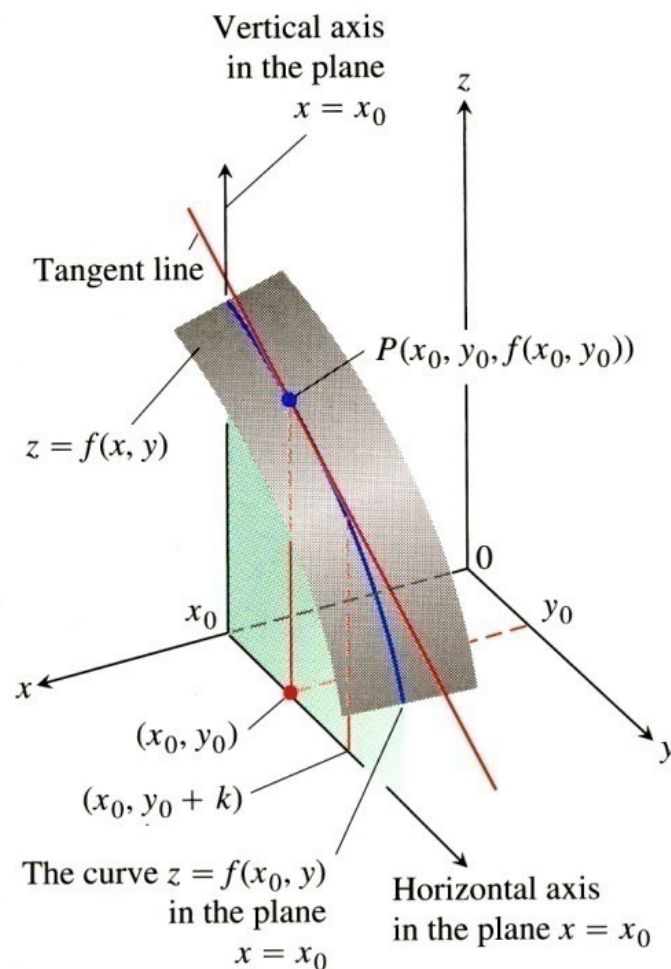
The symbol $\frac{\partial}{\partial x}$ is read as “partial derivative” with respect to x .

Partial Derivative of a Function of Two Variables

We can write an analogous expression for the partial derivative of $f(x, y)$ with respect to y .

The first Order Partial Derivative of a function $f(x, y)$ with respect to y holding x constant is then,

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \lim_{k \rightarrow 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k}$$



Other Notations for First Order Partial Derivatives

First order Partial Derivative of a function $f(x, y)$ with respect to x :

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)}, \quad f_x, \quad \frac{\partial z}{\partial x}, \quad z_x, \quad \left. \frac{\partial z}{\partial x} \right|_y, \quad \left. \frac{d}{dx} f(x, y_0) \right|_{x=x_0}$$

First order Partial Derivative of a function $f(x, y)$ with respect to y :

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)}, \quad f_y, \quad \frac{\partial z}{\partial y}, \quad z_y, \quad \left. \frac{\partial z}{\partial y} \right|_x, \quad \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0}$$



Example

If $f(x, y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial y}$ and $\frac{\partial f}{\partial x}$

Solution:

Higher Order Partial Derivatives

If $z = f(x, y)$ is a function of 2 variables, then its partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are also functions of 2 variables, and we can calculate their partial derivatives as,



Higher Order Partial Derivatives

Similarly,

These partial derivatives are called the **second order partial derivatives** of $f(x, y)$.

Higher Order Partial Derivatives

Example:

Calculate the second order partial derivatives of

$$f(x, y) = x \cos y + ye^x$$



Higher Order Partial Derivatives

Mixed Derivative (or Schwarz's) Theorem

If the second partial derivatives of $f(x, y)$ are continuous at (a, b) ,

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

Or, using our other notation,

$$f_{xy} = f_{yx}$$

These derivatives are referred to as mixed derivatives. While they may be denoted differently, they are equal in value.

Higher Order Partial Derivatives

Partial derivatives of order 3 can also be defined,

The mixed derivative theorem also applies for partial derivatives of order 3. For example,

And for functions of more than 3 variables,

The only requirement is to differentiate with respect to each variable the same number of times!



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Summary

- We learned the concept of partial differentiation
- We learned that to calculate partial derivatives we use the same set of rules for finding the derivative of a function of one variable (good news!)
- When we differentiate a function $f(x, y)$ twice, we produce second order partial derivatives.
- We have learned a second notation (other than the Leibnitz notation) to denote higher order partial derivatives.
- The mixed derivative (or Schwartz's) Theorem states that $f_{xy} = f_{yx}$ for continuous partial derivatives, and it also works for higher order partial derivatives
 - The only requirement is to differentiate with respect to each variable the same number of times!